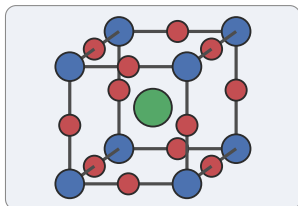


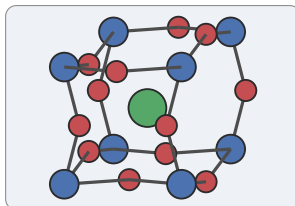
SrTiO₃

$Pm\bar{3}m$ · cubic perovskite
Goldschmidt $t = 0.95$



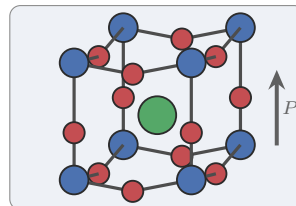
GdFeO₃

$Pnma$ · distorted perovskite
Goldschmidt $t = 0.80$



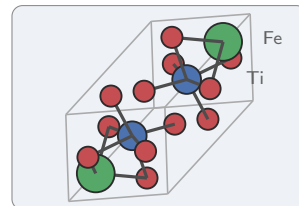
BaTiO₃

$P4mm$ · polar perovskite
Goldschmidt $t = 1.06$



FeTiO₃

$R\bar{3}$ · ilmenite
Goldschmidt $t = 0.70$



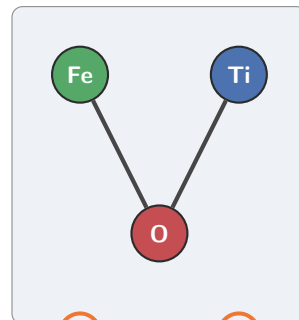
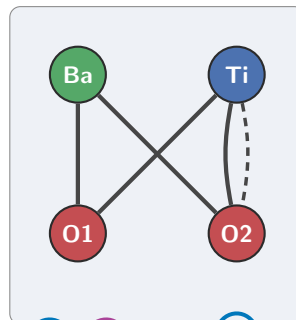
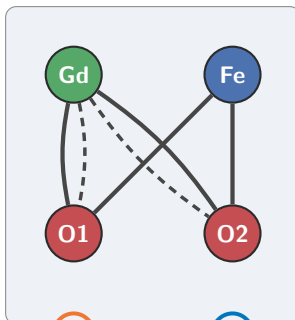
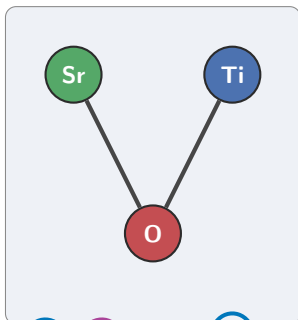
Structure

schematic projection

- A-site cation
- B-site cation
- oxygen

“Bonding” edges

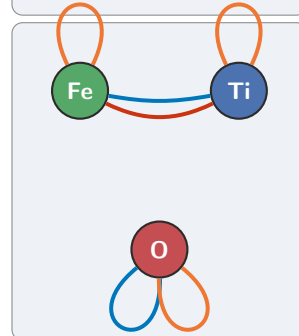
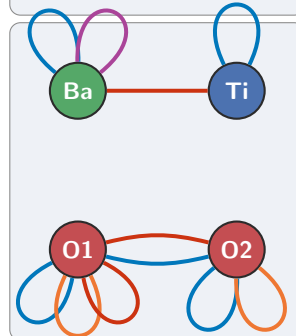
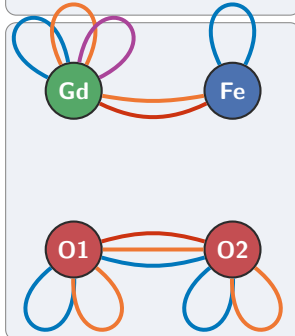
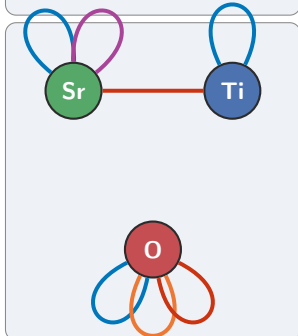
- core edge
 - - - extended edge
- one node per inequivalent site, one edge per symmetry class



Polyhedral edges

- corner sharing
- edge sharing
- face sharing
- 4+ sharing

loops = sharing between symmetry-equivalent sites



Topology similarity

	SrTiO ₃	LaNiO ₃	CaIrO ₃	FeTiO ₃
SrTiO ₃	—	0.947	0.885	0.231
LaNiO ₃	0.947	—	0.925	0.242
CaIrO ₃	0.885	0.925	—	0.265
FeTiO ₃	0.231	0.242	0.265	—

perovskites cluster:
topology ≥ 0.885
ilmenite apart:
topology ≤ 0.265

Distortion similarity

	SrTiO ₃	LaNiO ₃	CaIrO ₃	FeTiO ₃
SrTiO ₃	—	0.560	0.412	0.321
LaNiO ₃	0.560	—	0.642	0.349
CaIrO ₃	0.412	0.642	—	0.441
FeTiO ₃	0.321	0.349	0.441	—